01 / PROBLEM AND CONTRIBUTION

ESP32 GPS/GSM tracker PCB

Independent hardware design · KiCad | PCB design study

DESIGN AT A GLANCE

Integrated GPS, GSM, EEPROM, charging, and boost-converter sheets around an ESP32-S3 in a four-copper-layer KiCad tracker design.

Engineering question

How can a tracking board combine location, communication, local storage, and battery power?

Situation

The tracker architecture brings several interfaces and power blocks onto one board. The schematic identifies an ESP32-S3-WROOM-1, NEO-8Q GPS module, SIM800C modem, and CAT24C256 EEPROM.

Task

Organize the multi-sheet hardware design and board layout for the tracker and its power-management blocks.

My contribution

Schematic capture, PCB implementation, and design documentation in the linked personal GitHub project.

Tools and concepts

KiCad / Multilayer PCB layout / ESP32 hardware / UART interfaces / Power integration

02 / IMPLEMENTATION

Circuit to board

01 / Design step

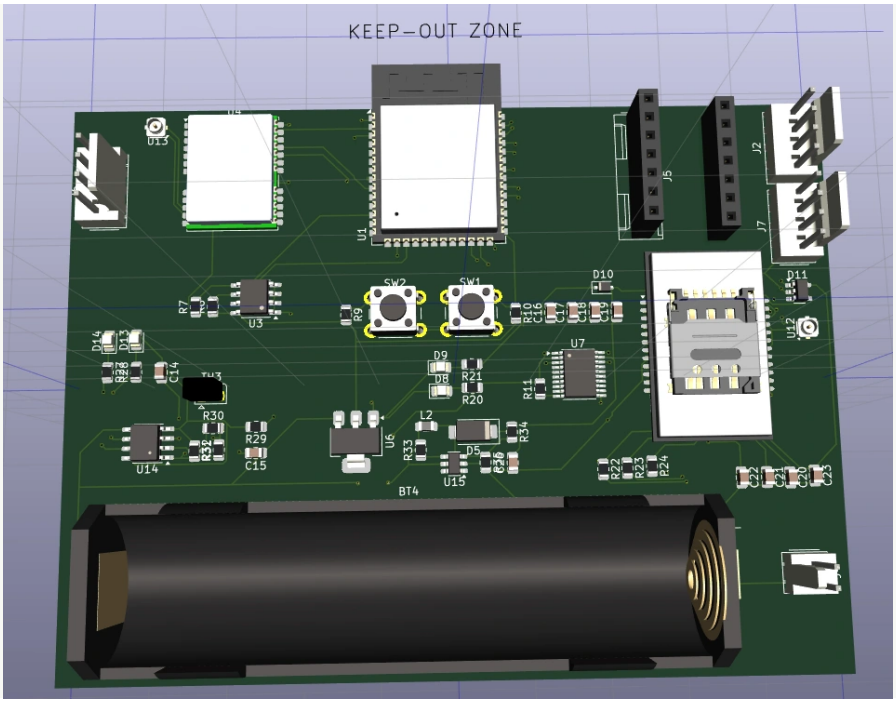
Separated GPS, GSM, connectors, charging, and boost circuitry into schematic sheets. The design includes TXS0108E level translation, TP4056 charging, and an MT3608 boost stage.

02 / Design step

Integrated the controller, local memory, external-module interfaces, and connectors in the PCB. The checked-in board contains four copper layers, 61 footprints, 770 track segments, and 84 vias.

03 / Design step

Documented the architecture and supplied a 3D board preview. Used the source files to distinguish intended tracking functionality from measured operation.



Tracker board render

Repository 3D render of the tracker board and battery holder.

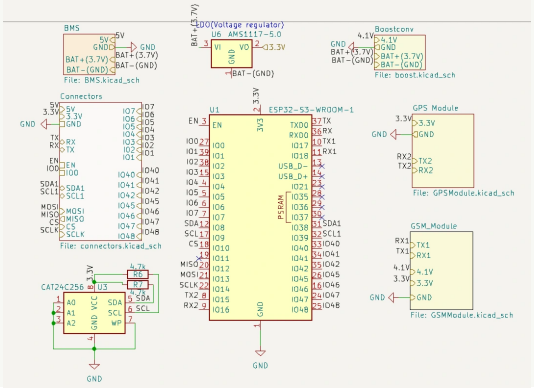
03 / EVIDENCE AND SCOPE

Design evidence and next steps

Current board	4 copper layers
Controller	ESP32-S3-WROOM-1
Position / cellular modules	NEO-8Q / SIM800C

Available evidence

The repository contains a multi-sheet schematic, routed board data, and a board render for the integrated tracker. These demonstrate the hardware-design work; no firmware logs, GPS-fix results, or cellular transmission measurements were found in the reviewed material.



Tracker top-level schematic

Repository architecture view linking controller, storage, power, GPS, and GSM sheets.

Scope and limitations

The schematic labels a regulator AMS1117-5.0 while the associated output net is labelled 3.3 V. This discrepancy needs resolution before assembly or power-up. No RF performance, impedance-control, battery-life, DRC-pass, or SI/PI sign-off claim is made.

Next steps - proposed work

- Resolve the regulator part-value and rail-label mismatch, then run ERC/DRC.
- Document firmware integration, GPS reception, cellular transfer, and supply transients.

Source: PCB-design-projects GitHub repository: GPS + GSM Tracker with EEPROM Logging; README, KiCad schematic/board, and repository previews. Reviewed 11 November 2026.

Open the project repository